

ADITYA RAJ

B.Tech CSE Student | Amrita Vishwa Vidyapeetham, Haridwar

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 [Portfolio](#)

EDUCATION

Amrita Vishwa Vidyapeetham, Haridwar

2025 – 2029

B.Tech – Computer Science & Engineering | CGPA: 8.0 (Semester 1)

Anjabi Singh College, Bihar Board

2023 – 2025

Class XII (Intermediate) | Percentage: 72%

The Divine Public School

Completed 2023

Class X | Percentage: 86%

TECHNICAL SKILLS

Languages

C, C++, Python, Java, JavaScript

Web Technologies

HTML, CSS

CS Fundamentals

Data Structures & Algorithms (DSA)

Tools & Platforms

Arduino, MATLAB, Simulink, Proteus, Tinkercad, NCviewer, NodeMCU (ESP8266)

Concepts

OOP, Graph Theory, Combinatorics, Discrete Mathematics, Mesh Analysis

PROJECTS

Personal Portfolio Website [\[Live Link \]](#)

2025–26

HTML / CSS / JavaScript | Electron.js

- ▶ Built a fully responsive personal portfolio with typing animation, cursor glow, scroll-reveal effects, and glassmorphism card design.
- ▶ Integrated 4 sub-projects: To-Do App, Quiz App, Shop, and Portfolio within a single Electron.js desktop application.

Smart Traffic Control System with Street Light

2025–26

Arduino UNO R4 WiFi | IoT | C++ / HTML

- ▶ Designed a 4-road smart intersection controller using Master-Slave Arduino architecture over WiFi (802.11 b/g/n).
- ▶ Integrated IR, Ultrasonic, LDR, and RFID sensors with SSD1306 OLED display and a REST HTTP API for real-time web control.

Autonomous Vehicle Navigation System – Graph Theory

2025–26

Discrete Mathematics | C++

- ▶ Modelled road networks as weighted directed graphs incorporating distance, traffic density, accident risk, and energy consumption.
- ▶ Applied Dijkstra, A*, and Kruskal algorithms to optimise travel time, congestion avoidance, and multi-vehicle coordination.

Automated Timetable Generator – Combinatorial & Recursive Techniques

2025–26

Discrete Mathematics | Python

- Formulated timetable scheduling as a combinatorial optimisation problem; solved via recursive backtracking.
- Eliminated teacher/room conflicts and met subject-frequency requirements for an entire academic semester automatically.

Electrical Circuit Mesh Analysis Using Inverse Matrix ($Ax = B$)

2025–26

Linear Algebra | MATLAB

- Applied KVL to multi-mesh circuits, converted simultaneous equations to matrix form, and solved using MATLAB's matrix-inverse method.

Library Management System Using OOP (Java)

2025–26

OOP | Java

- Implemented a Java-based LMS supporting book addition, member registration, and issue/return workflows using encapsulation, abstraction, and association.

Smart Light Detection & Alert System (LDR + NodeMCU)

2025–26

OOP | Java | IoT | ESP8266

- Built an LDR-based light intensity detector with LED and buzzer control on NodeMCU ESP8266, modelled in Java OOP for system logic.

Closed-Loop RPM Control System for DC Motor

2025–26

Manufacturing | Arduino | Proteus ISIS

- Designed a PID-controlled DC motor speed regulation system using Arduino UNO, 4N25 optocoupler, and L298 motor driver; simulated in Proteus ISIS.

MATLAB Simulation – Blackbody Radiation & Wien's Displacement Law

2025–26

Physics | MATLAB

- Simulated Planck's radiation law, plotted Intensity vs. Wavelength, and verified Wien's Displacement Law numerically across multiple temperatures.

CNC Rectangular Block Cutting – G-Code & M-Code Programming

2025–26

Manufacturing | CNC | NCviewer

- Programmed a complete CNC milling routine (ISO 6983) for a 100×60×10 mm aluminium block, verified with NCviewer toolpath simulation.

Pneumatic OR Logic Circuit (Shuttle Valve) – Simulink Simscape

2025–26

MATLAB/Simulink | Simscape Fluids

- Designed and simulated a dual push-button pneumatic OR logic circuit; validated pressure propagation and actuator response in Simscape Fluids.

INTERESTS & HOBBIES

Sports: Cricket, Badminton, Volleyball **Other:** Watching Movies, Travelling